Andrew McKnight

✉ andrew@mcknight.rocks
http://armcknight.com

Senior iOS developer with 14+ years of experience shipping developer tools, consumer apps and AI-integrated mobile experiences across all Apple platforms. Background spans low-level crash reporting and statistical profiling SDKs to user-facing apps in medical, eCommerce and B2B sectors, with full-stack exposure to backend systems. M.Sc. in Computer Science with published research in computational geometry and image processing.

Experience as both contractor and full-time employee, in office, remote and hybrid settings, including collaboration across timezones in North America and Europe.

iOS Experience

Jan 2021 – Current **Sentry**, *Senior iOS Developer*, Remote, (FTE).

- **Stack**: Objective-C++, Swift/SwiftUI, XcodeGen, GitHub Actions, SauceLabs, CocoaPods, SPM, Carthage, Swiftlint, clang-format, CodeCov, pre-commit, Fastlane, TestFlight, Arkana
- **Deliverable**: Sentry Cocoa SDK: mobile app crash and performance monitoring
- **Highlights**:
  - Spearheaded launching iOS/macOS SwiftUI apps for Seer AI workflows
  - Integrated Specto's profiler into Sentry's Cocoa SDK and iterated with new API/features
  - User Feedback widget: first user-facing UI shipped in Sentry Cocoa SDK
  - Built and maintained sample and demo apps for developers and Sales team
  - CI overhauls: added workflows, streamlined inconsistencies, and parallelized long-running jobs

Nov 2019 – Dec 2020 **Specto**, *Senior iOS Developer*, Remote, (FTE).

- **Stack**: Objective-C++, Swift, Go, Python, Jinja, Bazel, XcodeGen, GitHub Actions, pre-commit, clang-format
- **Deliverable**: Mobile app profiling SDK with shared C++ core between iOS and Android
- **Highlights**:
  - First employee at startup; grew and maintained iOS profiling SDK
  - Added safe init, launch profiling, signal propagation
  - Automated scoped Xcode project generation from Bazel targets (pre `rules_xcodeproj`)
  - Acquired by Sentry in December 2020

Jan 2019 – Oct 2019 **Rightpoint Consulting**, *Senior iOS Developer*, Remote, (Contract-to-Perm).

- **Stack**: Swift, SwiftGen, SwiftLint, Sourcery, CircleCI, Postman, Browserstack, Countly
- **Deliverables**: Security-focused app and SDK (NDA)

Jan – Apr 2018 **121nexus**, *iOS Developer*, Remote, (Contract).

- **Stack**: Swift, AVFoundation, OpenCV, Firebase, cocoapods-keys, AWS
- **Deliverables**: Medical inventory management app/SDK with barcode/QR scanning
- **Highlights**:
  - Replaced application UI and networking with their new SDK
  - Reimplemented major area of app with new UI and platform API
  - Added crash reporting (Crashlytics) and logging (XCGLogging)
  - Augmented and documented analytics (Firebase)
  - Assisted launches via iTunes (121 Scan) and another internal enterprise app

Dec 2016 – Jul 2017 **Layer**, *iOS Developer*, Remote, (FTE).

- **Stack**: C/Objective-C/Swift, CocoaPods, OCMock, Specta/Expecta, Jenkins, Rails, Thrift, WebSockets/SPDY, SQLite
- **Deliverables**: Objective-C SDKs for data and UI components, an embeddable app layer and fully-functional sample app
- **Highlights**:
  - Rearchitected test suite to separate unit tests from integration/end-to-end tests
  - Helped streamline developer environment configuration (Rake, Gems, CocoaPods and git submodules)
  - Migrated old SPDY/Thrift architecture to WebSockets with rewritten Thrift specs
  - Implemented new auth and transport layer using TransitionKit
  - Added SDK telemetrics implementation with SQLite

Apr 2015 – Dec 2016 **Twitter**, *Software Engineer*, Cambridge, MA, (FTE).

- **Stack**: C/Objective-C/Swift, Java/Scala/Clojure, Rails, React/Coffeescript, Storm/Heron, Cascading/Scalding, Graphite, Vertica/Zeppelin, Beanstalk/Kafka, Redis/Cassandra, AWS, TeamCity, Chef, Protobuf
- **Deliverables**: Crashlytics/Fabric SDKs, Mac app and command line tools, front- and back-end maintenance and analytics, Unity iOS support
- **Highlights**:
  - Streamlined build and release processes with TeamCity agents deployed via Chef
  - Added Bitcode and tvOS compatibility to SDKs
  - Standardized config of all Xcode projects and released the open source FABConfig
  - Maintenance of backend lambda architecture with on-call rotations
  - Pioneered new backend feature to cross-reference common crashes across apps and present developers with suggested fixes

Jan – Apr 2015 **MC10**, *iOS Developer*, Cambridge, MA, (FTE).

- **Stack**: Objective-C/Swift, CoreBluetooth, Android
- **Deliverable**: Companion iOS app for wearable medical peripheral with BTLE link
- **Highlights**:
  - Moved all app/demo code into Git repos (no previous SCM was used on the team)
  - Architected SDK and app codebases from the ground up
  - Built out greenfield app, in beta testing on my exit
  - Set up continuous integration and deployment with TeamCity and Fabric
  - Assisted adoption of Agile processes

Jul 2013 – Dec 2014 **Raizlabs Consulting**, *iOS Developer*, Boston, MA, (FTE).

- **Stack**: Objective-C/Swift, Jenkins, TeamCity
- **Deliverables**:
  - Care.com: service seeker/provider matching (4 months)
    - iOS 7 redesign, with perspective zoom effect for a horizontally paged scroll view
    - Dynamic UI generation from server-supplied data
  - B&H Photo: e-commerce (5 months)
    - Custom parallactic scrolling of table view headers
    - Push notifications with deep linking
  - UniFirst: interactive enterprise sales catalog on iPad (8 months)
    - Drag and drop editing of clothing selections, with PDF report generation
  - Invaluable: realtime high-end auction platform on iPad (1 month)
    - Paging of API result display in table/collection views

Oct 2012 – May 2013 **ODU Systems Lab**, *R&D projects*, Norfolk, VA, (Student Intern).

- **Deliverables**:
  - ChirpXM: location based social networking app
  - EMR iPad app for CHKD
- **Highlights**:
  - Supervised two foreign exchange students
  - Regular consultations with onsite clients

Aug 2012 – May 2013 **Old Dominion University**, *Undergraduate capstone project*, Norfolk, VA, (Seminar).

- **Stack**: Objective-C, OpenEars, MapQuest
- **Deliverable**: Traffic Wizard, a mobile app to help drivers avoid traffic jams
- **Highlights**:
  - Co-lead of 7 member team
  - 4 presentations given to visiting professors and industry executives
  - Sole developer of the team's iOS portion
  - Highest scoring project!

Nov 2011 – Current **Insight Therapeutics**, *Lead iOS Developer*, Norfolk, VA & Remote, (Contract).

- **Stack**: Objective-C/Swift, MessageUI, Storyboards, CoreGraphics, CoreData, FMDB, SQLite, SSZipArchive, RNCryptor, cocoapods-keys
- **Deliverables**:
  - ClotRx & DementiaRx: reference e-pamphlets
    - Hybrid app with WKWebView
    - Searchable, bookmarkable HTML content automatically extracted from PDFs with custom scripts
    - Interactive clinical surveys, print/email report generation
  - Antibiotic Kinetics: antibiotic dosage calculator
    - Ported from another developer's preexisting Android app
  - eConsent: HIPAA-compliant informed consent app for clinical trials
    - Interactive quiz-based presentation of study details
    - Signature collection and export to PDF
    - Interface with backend API to upload finished consents

Teaching and Presentations

May 2018 **iOSDevCampCO**, *Colloquial Swift (45 min)*, Colorado Springs, CO.

- Presented findings from Colloquial Swift personal project (see entry in previous section)

Jun 2017 **CocoaHeads Boston**, *UIViewController Decomposition and UI Reuse (20 min)*, MIT, Cambridge, MA.

- Discussing strategies to separate responsibilities from UIViewControllers and a library of reusable UI components, e.g. animated transparent modal presentation with a close button

Jan 2017 **Boston Algorithms Meetup**, *HyperLogLog (20 min)*, Twitter, Cambridge, MA.

- Helped organized this initial meeting of the group
- Covered the theoretical workings of the HyperLogLog algorithm, with a copresenter giving a working demostration

Jul 2016 **Girls Who Code**, *iOS Dev Session*, Twitter, Cambridge, MA.

- Part of GWC summer internship covering programming first principles to full-stack style development
- Designed and led iOS session where we built a Flappy Bird clone, introducing them to Xcode as an IDE, Interface Builder, Playgrounds and Swift

Mar 2016 **mDevCon**, *Command and NSOperation (45 min)*, Amsterdam, NL.

- Delivered presentation on the Command software design pattern, and NSOperation
- Introduced subclasses we open sourced at Crashlytics for asynchronous and compound operations
- Previously delivered at a lunchtime tech talk at Twitter, Cambridge, MA (Nov 2015) and CocoaHeads Boston meetup, MIT, Cambridge, MA (Dec 2015)

Oct 2015 **NeighborNest**, *Web Dev Session*, Twitter, San Francisco, CA.

- Community program to help advocate technology literacy
- Assisted in session where students worked on a Twitter clone, teaching functional programming concepts, Javascript, and some front- and back- end concepts